Electronic Devices and Circuits Lab

8-HOUR DIGITAL CLOCK



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Contents

Contents **1**

1 Digital Clock **2**

1.1 Introduction 2

1.2 Components 2

1.3 Working Principle 2

1.3.1 Pulse Generation (Arduino UNO) 2

1.3.2 Counting Mechanism 3

1.3.3 BCD to 7-Segment Decoding 5

1.3.4 Resetting using Gates 5

1.4 Block Diagram of the Circuit 6

1.5 Results 6

1.5.1 Seconds Display 7

1.5.2 Minutes Display 7

1.5.3 Hours Display 7

1.6 Conclusion 7

1.7 Precautions 8

1

Digital Clock

1.1 Introduction

This experiment aims to design and demonstrate the digital clock using Arduino for clock generation, seven-segment displays, D flip-flop counters, decoders and some logic gates to reset the counters. The whole project is operated by using the 1Hz frequency generated by Arduino to give it as the CLK to the initial counter and count the seconds, minutes and hours in this 8-hour digital clock.

The main objective is to demonstrate the functional digital clock that displays the seconds, minutes and hours.

1.2 Components

i) Arduino Uno
ii) Five 7-segment displays
iii) SN74LS47N BCD to 7-segment decoder ICs
iv) SN74LS74N D flip-flop ICs
v) 7400 NAND gate ICs
vi) 7404 NOT gate ICs
vii) Connecting and Jumper wires

1.3 Working Principle

1.3.1 Pulse Generation (Arduino UNO)

To give the clock to the circuit, we are using the pulse of 1Hz frequency generated by the Arduino UNO. This pulse is used in the MOD-10 counter of seconds to increment it. The Arduino Uno generates a 1 Hz pulse, which serves as the clock signal for the entire circuit. This pulse is used to increment the seconds counter in the clock.

1.3.2 Counting Mechanism

1. **Seconds Counter** :
   The seconds counter (SN74LS74N D flip-flop IC) comprises a MOD 10 counter and a MOD 6 counter to count from 00 to 59 seconds of the clock. A 1Hz pulse is given by the Arduino and is fed into the first MOD-10 counter of the seconds counter, which then upon reaching the 9 sends out a carry-out pulse to the MOD-6 counter. Once the seconds counter reaches the 59 seconds mark, the counter resets the number to 00 and sends the carry-out pulse to the minutes counter.
2. **Minutes Counter** :
   The carry-out pulse given by the Seconds counter is connected to the Minutes counter as a clock to increment the Minutes counter, which also comprises MOD-10 and MOD-6 counters using the SN74LS74N D flip-flops. This also works similarly to the seconds counter after reaching 59 from 00 it resets at 00 while sending the carry-out pulse to the hours counter.
3. **Hours Counter** : The carry-out pulse given by the Minutes counter is connected to the Hours counter as a clock to increment the Hours counter, which comprises MOD-8 counters using the SN74LS74N D flip-flops. This counter also works similarly to the other two counters but this ranges only from 00 to 07. After reaching the seven it resets the counter to 00.

Below Fig. 1.1 is the MOD-10 counter used in the seconds and minutes counter:

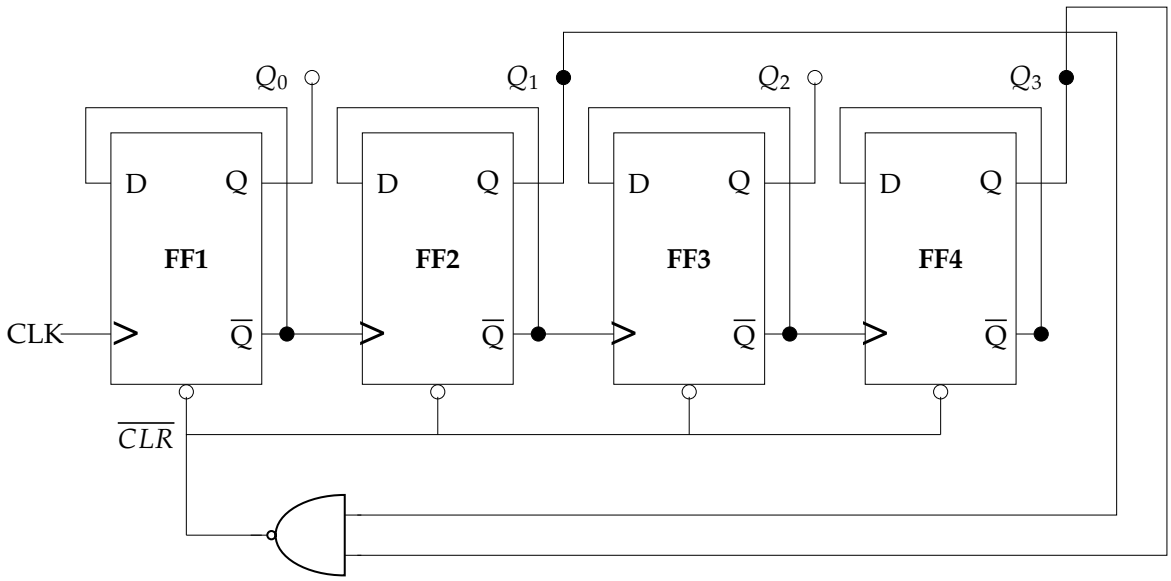


Figure 1.1: *Modulo-10 Counter Circuit*

Below Fig. 1.2 is the MOD-6 counter used in the seconds and minutes counter:

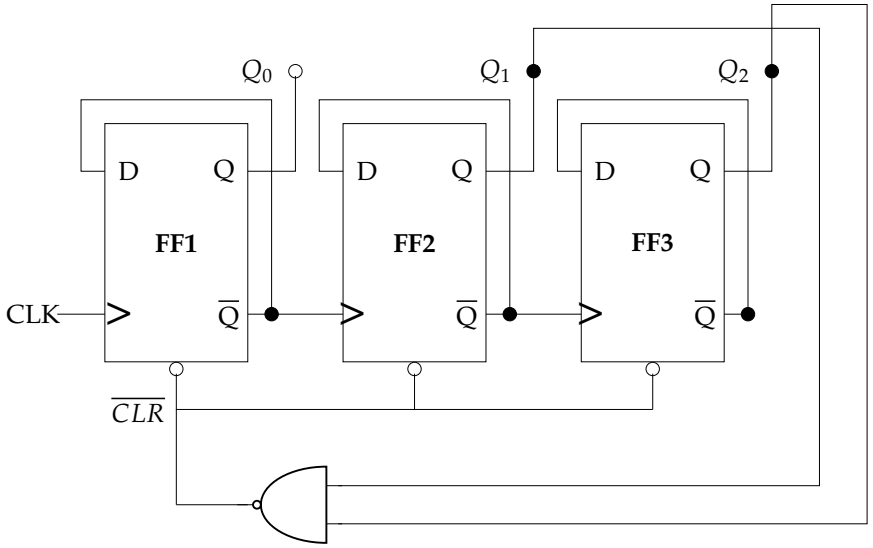


Figure 1.2: *Modulo-6 Counter Circuit*

Below Fig. 1.3 is the MOD-8 counter used in the hours counter:

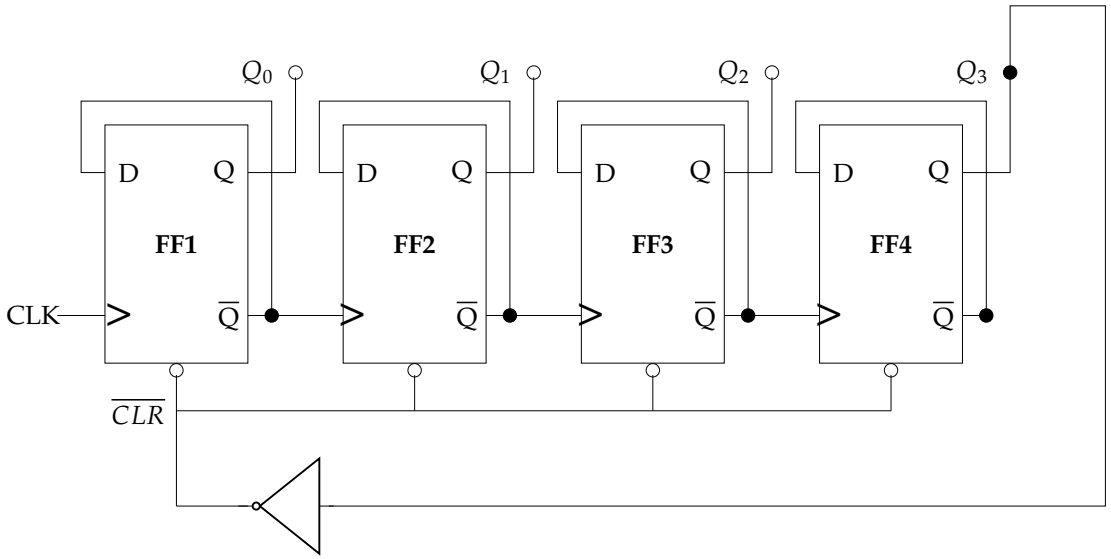


Figure 1.3: *Modulo-8 Counter Circuit*

Below is the truth table of a D-flipflop which represents how a D-FlipFlop works according to the clock and the reset. Here reset is used to keep the flipflop in a known state regardless of the input.

Reset (R)	Clock (CLK)	D (Input)	Q (Output)	Q′ (Complement)	Description
1	X	X	0	1	Asynchronous Reset
0	↑	0	0	1	Q follows D
0	↑	1	1	0	Q follows D
0	0	X	Q_{prev}	Q'_{prev}	No change

Table 1.1: *Truth Table of a D Flip-Flop with Reset and Clock*

1.3.3 BCD to 7-Segment Decoding

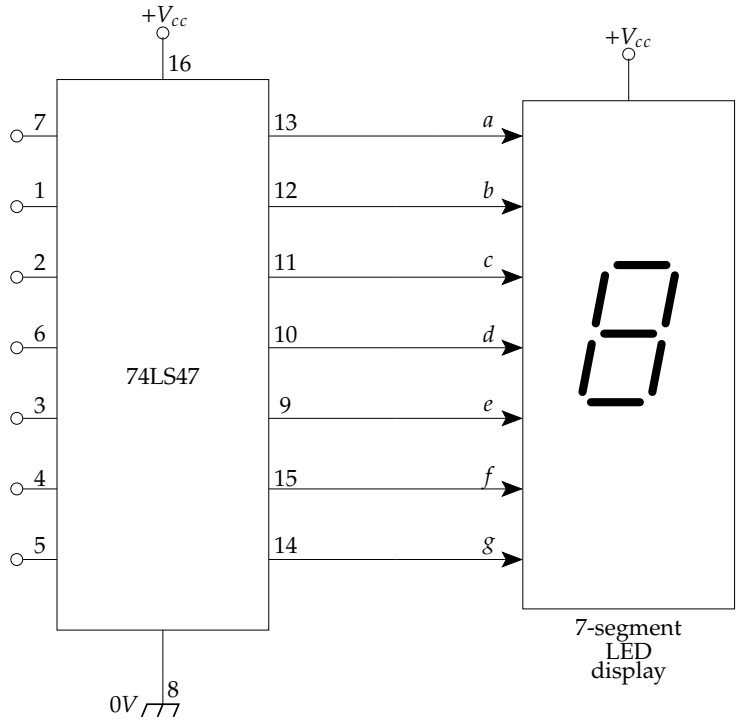


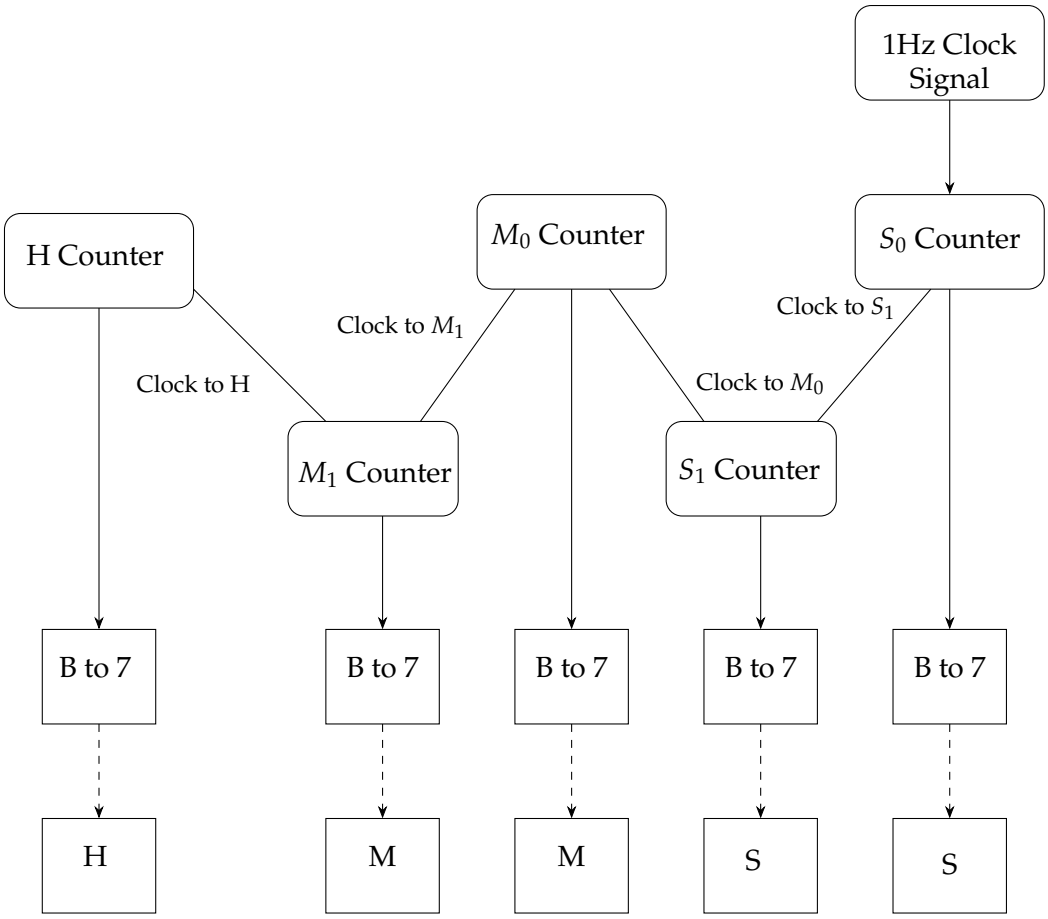
Figure 1.4: *BCD to 7-segment decoding*

The BCD (Binary-Coded Decimal) output from each counter is given to the corresponding SN74LS47N BCD to the 7-segment decoder IC. This decoder is used to convert the BCD data into a form which is readable for the 7-segment displays, to display the digits (00 to 07 for hours, and 00 to 59 minutes and seconds). The five 7-segment displays are connected to the output of these decoders where two displays represent the seconds, the other two minutes and the remaining one the hours.

1.3.4 Resetting using Gates

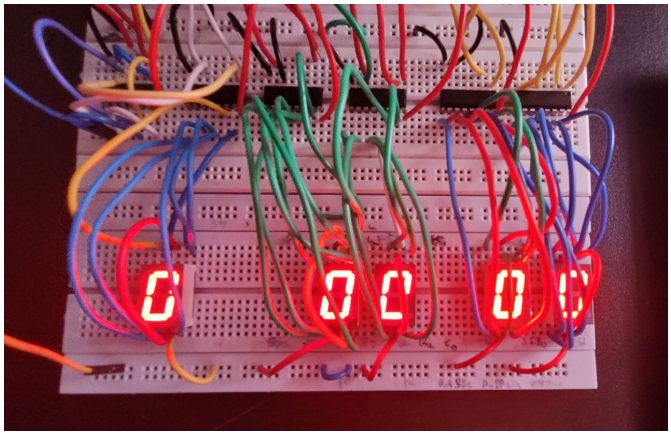
1. **7400 NAND Gate ICs** : These NAND gates are used to reset the hours, minutes and seconds without crossing the range of the desired values. Here we are supposed to give two inputs and the corresponding output is given to the $\overline{CLR}$ pins to reset the FlipFlop according to a certain logic.
2. **7404 NOT Gate ICs**: These NOT gates are also used to reset the hours without crossing the range of the desired values. For example, in the hours it should range from 0000 to 0111, so we need to reset the counter at 1000 we use a NOT gate to reset to 0000 when the MSB hits logic '1' by triggering the $\overline{CLR}$ pin.

1.4 Block Diagram of the Circuit



1.5 Results

Below are the images captured during the experiment :



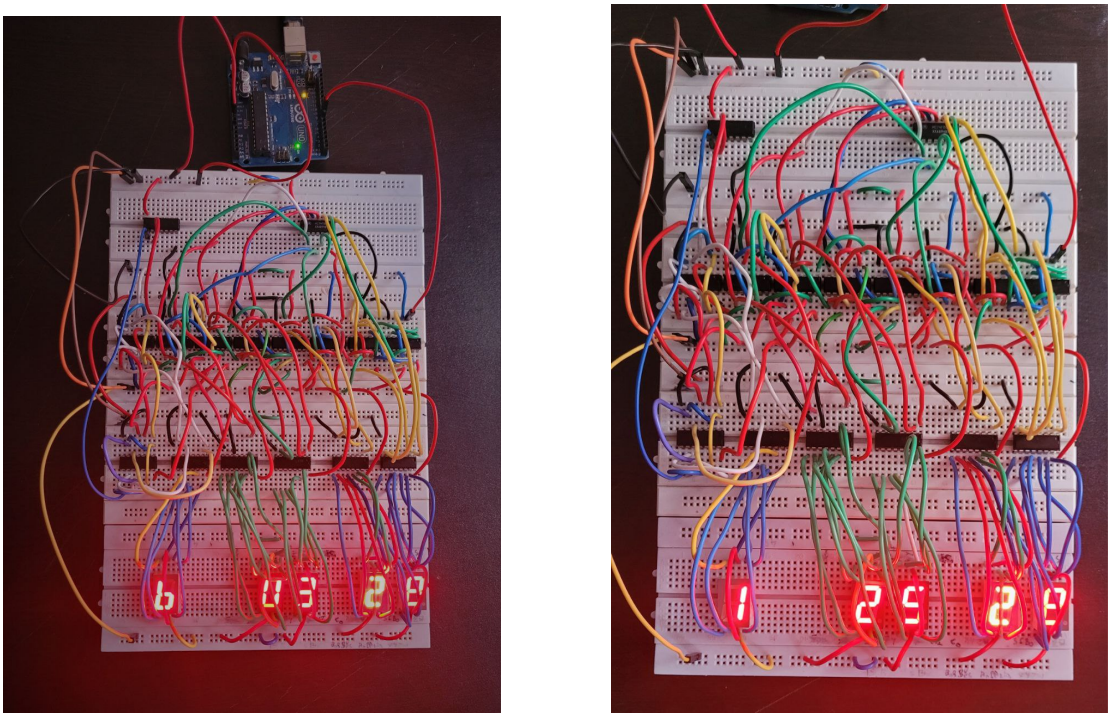


Figure 1.5: *Experimental setup for the 8-hour digital clock project*

1.5.1 Seconds Display

- **Units of Seconds (0-9)**: increments correctly from 0 to 9 and rolls over after 9 to 0.
- **Tens of Seconds (0-5)**: increments correctly from 0 to 5 and rolls over after 5 to 0 (when the unit digit goes back to 0).

1.5.2 Minutes Display

- **Units of Minutes (0-9)**: increments correctly from 0 to 9 and rolls over after 9 to 0.
- **Tens of Minutes (0-5)**: increments correctly from 0 to 5 and rolls over after 5 to 0 (when the units digit goes back to 0).

1.5.3 Hours Display

- **Hours (0-7)**: increments correctly from 0 to 7 and rolls over after 7 to 0.

1.6 Conclusion

In the above experiment, we have successfully demonstrated an 8-hour digital Clock using the Counters of D-FlipFlops, Arduino for 1Hz frequency, 7-segment decoders and displays where we achieved a working digital clock that accurately displays the time with the help of 7-segment displays with the help of pulse generated by Arduino. With the help of this project, we understood the mechanism of counters and combinational logic circuits.

1.7 Precautions

- Ensure that all the counters are properly initialised.
- Ensure that no component is overloaded beyond their rated values.
- Ensure that all the wirings are made according to the schematic diagram, as miswiring may damage the component.